

Rulebook Drift Monitor

A governance loop for financial-crime detection rulebooks. LLM agents red-team your typology playbook, quantify how well your indicator rules still catch threats, and let an analyst institute new rules that measurably close the gaps. Every action screens live risk of threats in a portfolio of regions.

Rulebook Drift Monitor

Core concepts

a model risk & governance platform for financial-crime detection rulebooks

- Drift index (0-100): how far the live rulebook has drifted from the risk baseline. 40+ is elevated, 70+ is red.
- Typology x category matrix: 35 red-team typologies (customer, product, delivery, geography...) against 14 rule categories.

Everything you see on the red team reproduced on real fixtures - an LLM drafts candidate red flags, the rule engine verifies them.

Interactive showcase guide

every page of the app explained - plus a 30-minute demo script and quick-start commands

- The amendment loop: approve a finding and institute it as a new dedicated rule (DS-41+); the engine self-checks the new rule and the covered typology drops out of the drift measurement.
- Diffable & exportable: each run is diffed against the previous (NEW / REPEAT / REGRESSED) and a compliance dossier is one click away (JSON + PDF).

Live state of this instance (at a glance)

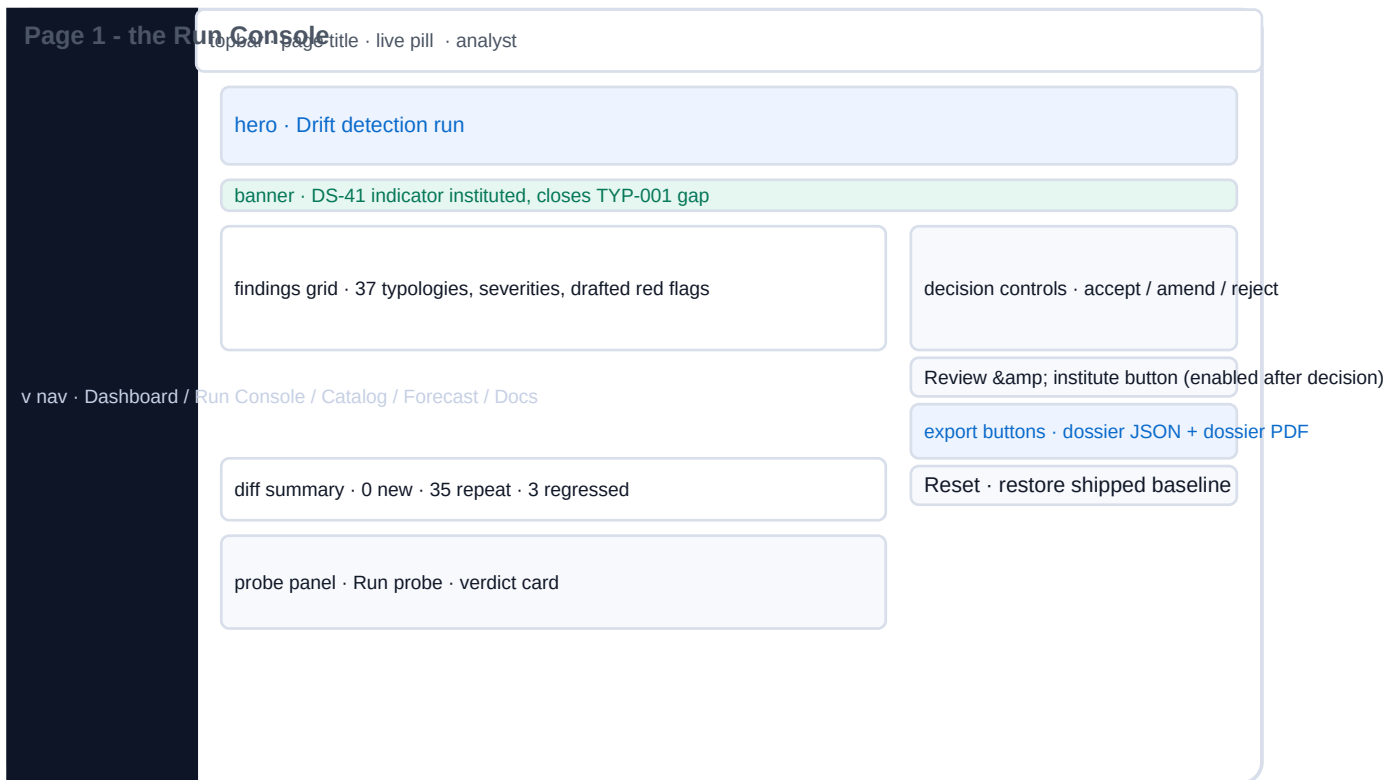
Metric	Value
Locked baseline rulebook	40 rules (DS-01..DS-40), restored on Reset
Latest completed run	run-b39c5c27 - 37 verified findings awaiting human approval
Drift now	52.9 (down from 54.6 baseline) after DS-41 institutionalised on TYP-001

How to run it yourself

- `python3 demo/web.py` then open `http://127.0.0.1:5000`
- New scan on the Run Console, review, institute a rule, watch the dashboard & forecast react, export the dossier.

Page 1 - the Run Console (index.html)

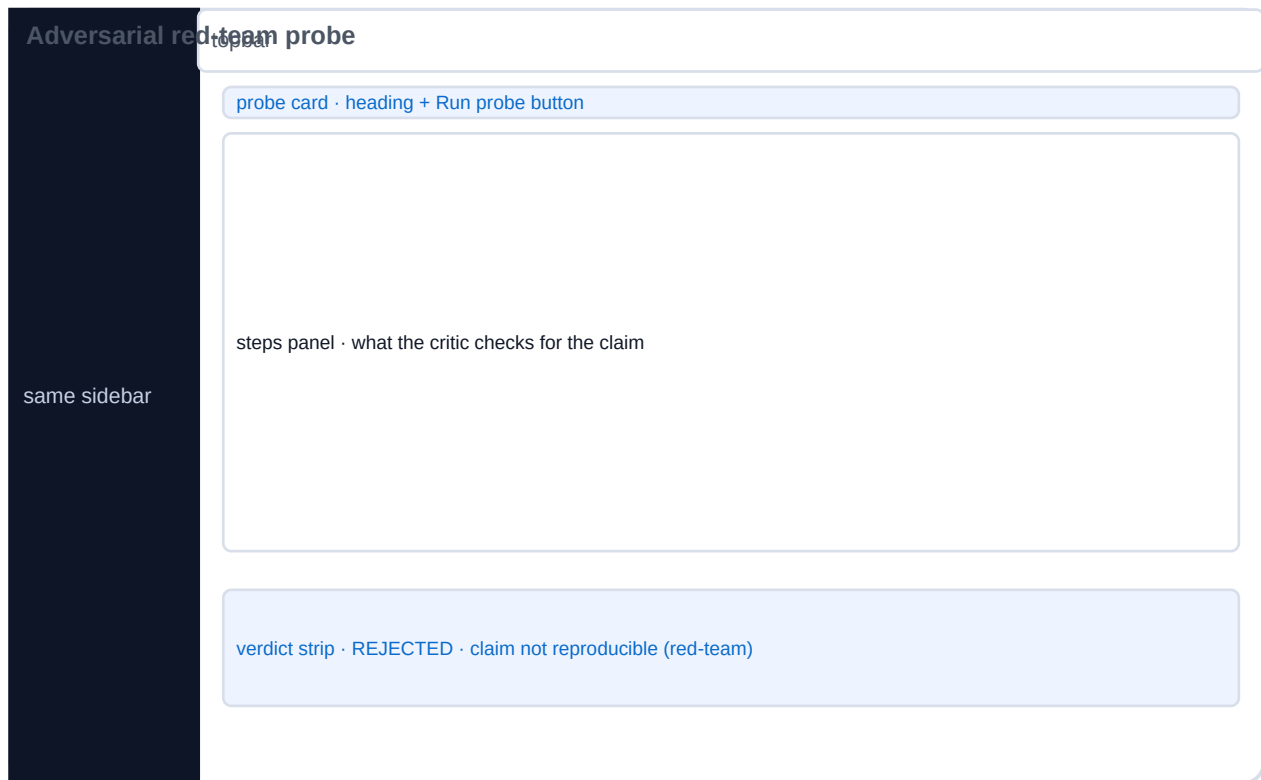
Command centre: start scans, review findings, institute rules.



- **New run button** starts a fresh red-team scan; the pipeline trace (retrieval -> drafting -> feasibility -> rule engine) goes live as it runs.
- **Findings grid** - one finding per typology: typology, category, severity, evasion depth and its drafted red flag.
- **Decision controls** - accept / amend / reject set the status; **Review & institute** commits it as a new rule.
- **Diff badges** - NEW (first seen) vs REPEAT, plus the summary line against the previous run.
- **Run toggles** - Reset, Export dossier JSON and Export dossier PDF.
- **Institutionalised banner** - appears once rules are in force (e.g. DS-41 · closes TYP-001).

Adversarial red-team probe (Run probe)

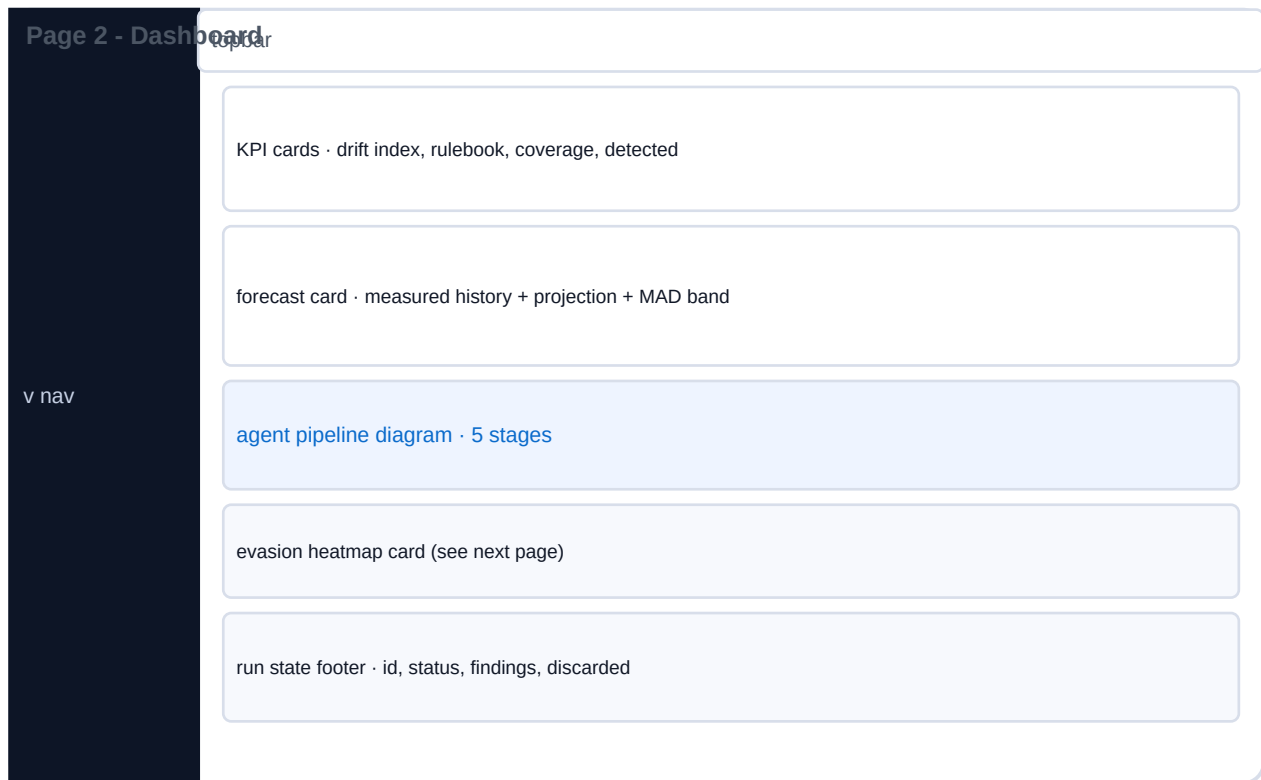
A live attack simulation built into the console.



- **How it works** - hands the platform a poisoned typology: a prompt-injection payload plus an over-claimed evasion set.
- **Verification** - the engine re-runs feasibility checks; the steps panel shows exactly what was tested.
- **Verdict** - REJECTED when the over-claim does not reproduce; honestly FLAGGED if the injection truly bypasses the stack.
- **Live use** - one click, ~1 second, unambiguous result for the security story.

Page 2 - Dashboard (dashboard.html)

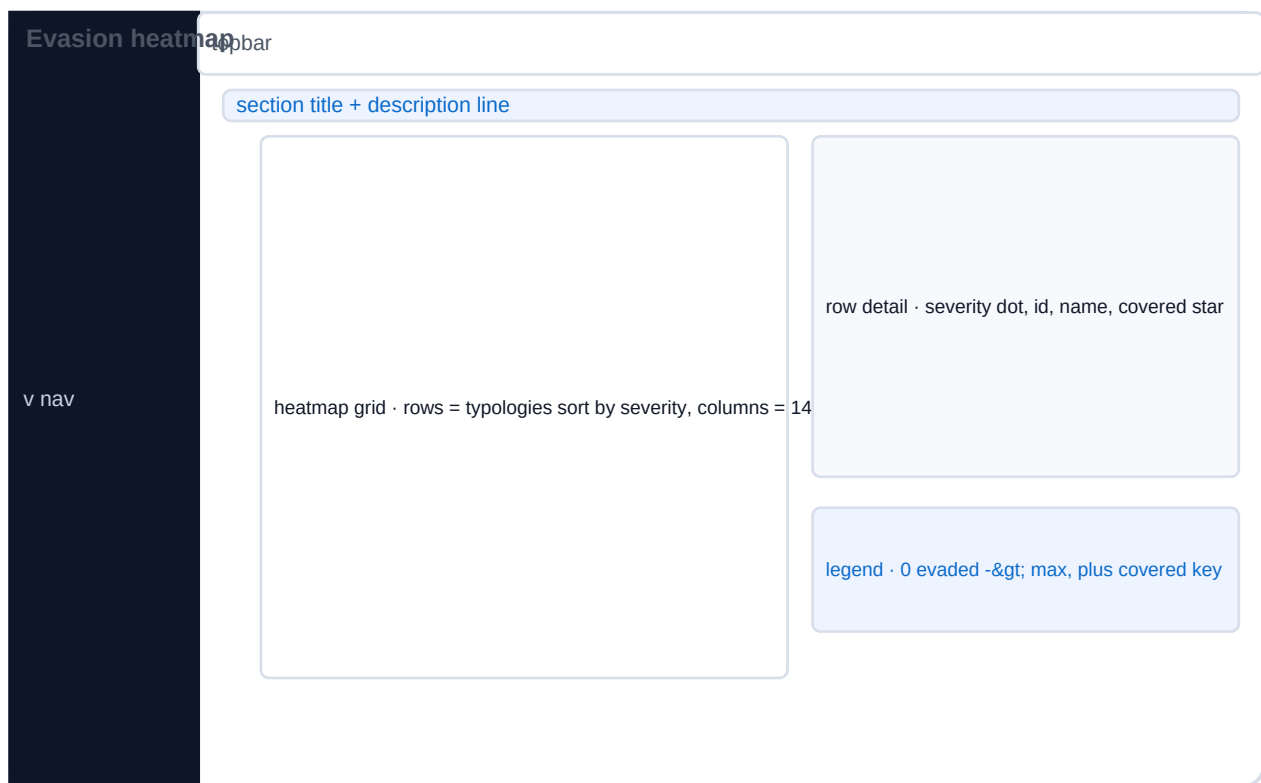
The 100-metre view: KPIs, forecast, pipeline, current run.



- **KPI cards** - drift index (with a pill "+ 1 typology gap closed"), rulebook size, rule coverage %, detected typologies.
- **Forecast card** - measured history line, projected +75.8 drift in 30 days and a backtested error band (MAD).
- **Pipeline diagram** - the five agent stages light up as a run progresses.
- **Run state footer** - current run id, status, findings and discarded counts.

Evasion heatmap (dashboard)

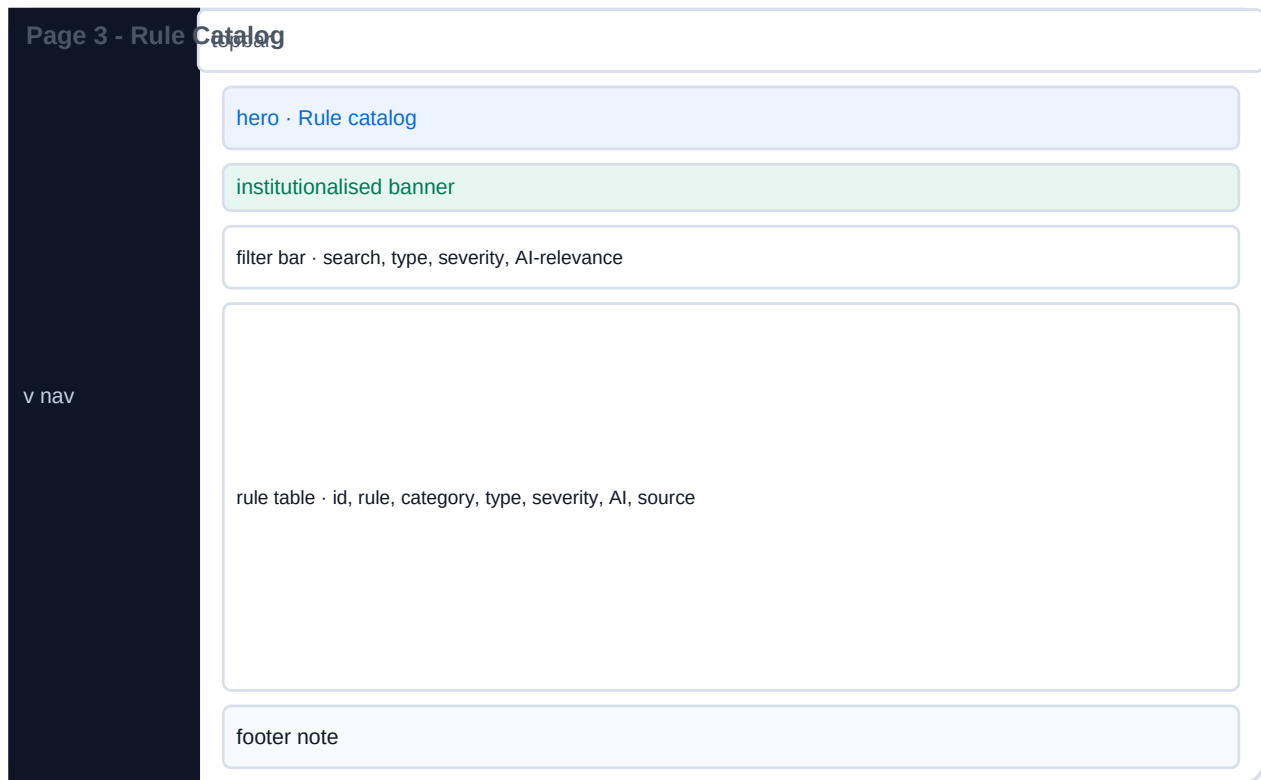
Typology x category evasion, at a glance.



- **Rows** are typologies sorted by severity (critical first) then total evasion; a severity dot sits on each row label.
- **Columns** are the 14 rule categories; intensity = number of evaded rules, printed inside the cell.
- **Right-most column** - total rules evaded per typology.
- **Green outline + star** - a covered typology is excluded from the drift measurement (DS-41 neutralises TYP-001).
- **Legend** - intensity scale and the covered key are drawn in the chart itself.

Page 3 - Rule Catalog (rules.html)

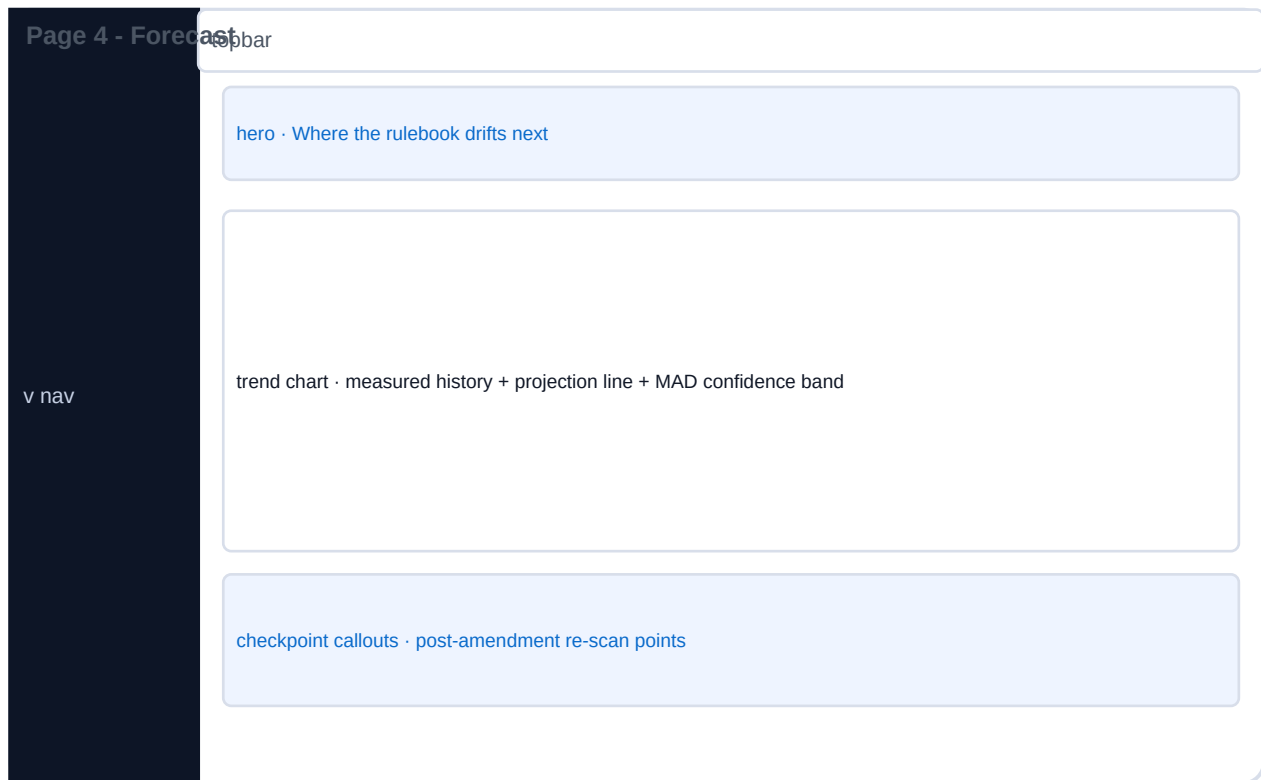
The rulebook, browseable and auditable.



- **40 baseline rules** grouped by category - rule id, title, category, risk severity, conditions.
- **Star badge (instituted)** - a rule created through the amendment loop shows which finding it closes.
- **Filter/search** - narrow by text across id/title/conditions.
- **Audit story** - the catalog always reflects what is actually in force.

Page 4 - Forecast (forecast.html)

Where the drift is heading, with receipts.



- **Measured history** - seeded regulatory steady-state points plus your completed runs (currently 9 points).
- **Projection** - trend line to +30 days (~75.8): the board-facing number.
- **Backtest** - MAD / confidence band showing past-model accuracy.
- **Checkpoints** - every amendment appends a measured point, so the curve visibly bends when the analyst acts.

The amendment loop - the story to tell

Stage	What happens	Where you see it
1 · Scan	Agent red team drafts a candidate red flag; the rule engine verifies it reproduces.	Run Console grid
2 · Review	Analyst accepts the finding; Review & institute is now enabled.	Finding card
3 · Institute	Engine builds a dedicated DS-rule, self-checks it, commits it, marks the typology covered.	Finding card + banner
4 · Measure	Post-amendment re-scan records a new drift point; the forecast recalibrates.	Dashboard + Forecast
5 · Re-run	Next scan excludes the covered typology; its findings come back REPEAT / REGRESSED.	Run Console diff
6 · Export	Dossier (JSON + PDF) assembles approved findings and amendments.	Export buttons

The 30-minute demo script

Time	Action
0:00 Check-in	Show the live pill and the running server; open http://127.0.0.1:5000 .
0:02 The problem	One sentence: "Your indicator rulebook silently rots as real-world typologies change - we make that drift visible and closeable."
0:03 Run Console	Point at a finding card. Accept TYP-001, then Review & institute.
0:06 The loop	Read the banner back: DS-41 instituted, TYP-001 gap closed; drift KPIs drop (54.6 -> 52.9). Differentiator.
0:09 Probe	Click Run probe; "REJECTED - not reproducible" in under a second.
0:12 Dashboard	KPIs, then the heatmap: severity-sorted rows, green-outlined covered TYP-001, counts in cells.
0:15 Forecast	The measured line + projection; the amendment checkpoint bends the curve.
0:18 Re-run	New run (fast - cached drafting); findings return REPEAT with the summary diff.
0:22 Export	Export dossier (JSON) and PDF; open it and show amendments + approved findings.
0:25 Reset + Q&A	Reset restores the pristine baseline (40 rules, clean history). Q&A.
0:28 Wrap	"From a live red-team red flag to an instituted, measured, exportable rule in four clicks - that is the drift monitor."

Quick references

- `python3 demo/web.py` - start the app on port 5000 (all data synthetic / illustrative).
- Pages: `/index.html` Run Console, `/dashboard.html`, `/rules.html`, `/forecast.html`, `/docs.html`.
- APIs: `/api/run` `/reset` `/decide` `/institute` `/probe` `/dossier` `/dossier.pdf` `/dashboard` `/forecast` `/rules` `/state`.